Meta Ads: Ad Performance Analysis

1. Project Overview

This project analyzes Meta advertising performance on Facebook and Instagram using data from 50 campaigns, 200 ads and 400,000 user-level ad events recorded between 7 May and 6 August 2025. The goal is to give the marketing team visibility into campaign reach, engagement, conversions and budget use, so they can identify the most effective platform, track campaign return, optimize budget allocation and understand audience engagement patterns.

The scope covers paid campaigns on Facebook and Instagram only. Other platforms (Messenger, Audience Network) and organic engagement are out of scope.

2. Dataset Summary

- Files: 4 – ads (200 rows, 7 columns), campaigns (50 rows, 6 columns), users (9,841 rows, 7 columns) and ad_events (400,000 rows, 7 columns)
- Key Features:
- Ad details (Platform, Ad Type, Target Gender, Target Age Group, Target Interests)
- Campaign details (Name, Start Date, End Date, Duration, Total Budget)
- User demographics (Gender, Age, Age Group, Country, Location, Interests)
- Ad events (Timestamp, Day of Week, Time of Day, Event Type: Impression, Click, Like, Comment, Share, Purchase)
- Ads: 127 Facebook and 73 Instagram; Stories 64, Image 52, Carousel 51, Video 33
- Events: Impression 339,812 (85.0%), Click 40,079, Like 12,013, Comment 4,108, Purchase 2,031, Share 1,957
- Event Period: 7 May 2025 to 6 August 2025
- Missing Data: none – no missing values or duplicate rows in any of the four files

3. Data Cleaning and Preparation using Power Query

We cleaned and prepared the four files in Power Query before loading them into the Power BI data model:

- **Data Loading:** Imported ads.csv, campaigns.csv, users.csv and ad_events.csv (200, 50, 9,841 and 400,000 rows) and checked column profiles.

- **Data Type Standardization:** Set user_id to Text so identifiers are never read as numbers, converted campaign start_date and end_date from dd-mm-yyyy text to Date, and converted the event timestamp to Date/Time (all 50 campaign dates and 400,000 timestamps valid).

- **Missing and Duplicate Values:** Checked all four tables for nulls, duplicate rows and duplicate keys (event_id, ad_id, campaign_id, user_id); none were found, so no rows were removed. Text columns were trimmed and categorical values checked; no changes were needed.

- **Consistency Checks:** Verified that duration_days equals end date minus start date for all 50 campaigns, that day_of_week and time_of_day match the event timestamp for all events, and that user age groups match user ages; no mismatches were found.

- **User ID Standardization:** In users.csv, 225 user IDs were stored in scientific notation (for example 1.20E+01) and 117 had lost their leading zeros, so 342 of 9,841 IDs (3.5%) no longer matched ad_events, and a direct join left 20,046 events (5.0%) without a user. A standardized key was created in both tables by converting numeric-looking IDs to a number and back to text and keeping other IDs as text. This recovered 11,002 events, raising linked events from 379,954 (95.0%) to 390,956 (97.7%).

- **Unresolved User IDs:** Two standardized keys are shared by 64 different users, so those users were excluded from the users table (9,777 remain) and their 2,917 events (0.7%) were labeled Unknown. A further 6,127 events (1.5%) have no matching user in users.csv. No events were deleted, so the 9,044 Unknown events (2.3%) stay in all KPI totals but are left out of the country analysis.

- **Campaigns Without Ads:** Campaigns 16 and 43 have no ads. They were kept (their $152,872, 6.0% of the total budget, stays in Total Budget and Avg Budget per Campaign) and flagged with a has_ads column so they can be excluded from the budget-efficiency analysis.

- **Campaign Dates vs. Event Dates:** Only 44.4% of events fall inside their campaign's start and end dates, so campaign dates were kept as descriptive fields and all time-based views use the event timestamp.

- **Feature Engineering:** Added event hour (0–23) and event date from the timestamp, a Date table with week number and month for the weekly trend and calendar heat map, and the standardized user key.

- **Data Model:** ad_events link to ads (ad_id), users (standardized user key) and the Date table, and ads link to campaigns (campaign_id). The 12,013 Like events are kept but excluded from Engagements, as defined in the BRD (Clicks + Shares + Comments).

4. KPIs and Business Analysis

We calculated the KPIs defined in the BRD and analyzed performance by platform, ad type, audience, country, time and budget. Differences in CTR, conversion rate and purchase rate between groups were tested with chi-square tests of independence on event counts (for example, clicks vs. impressions without a click) at a 5% significance level, and relationships between budget and results are Pearson correlations across the 48 campaigns with ads. A difference called significant passed this test; all others did not.

1. **Overall KPIs** – Calculated the BRD KPIs across all campaigns.

KPI	Formula	Overall
Impressions	Count of Impression events	339,812
Clicks	Count of Click events	40,079
Shares	Count of Share events	1,957
Comments	Count of Comment events	4,108
Purchases	Count of Purchase events	2,031
Engagements	Clicks + Shares + Comments	46,144
CTR	Clicks ÷ Impressions × 100	11.79%
Engagement Rate	Engagements ÷ Impressions × 100	13.58%
Conversion Rate	Purchases ÷ Clicks × 100	5.07%
Purchase Rate	Purchases ÷ Impressions × 100	0.60%
Total Budget	Sum of campaign budgets (50 campaigns)	$2,535,924
Avg Budget per Campaign	Total Budget ÷ Campaign Count	$50,718

2. **Platform Performance** – Compared Facebook and Instagram on reach, engagement and conversion.

Metric	Facebook	Instagram
Ads	127	73
Impressions	215,972	123,840
Clicks	25,389	14,690
Purchases	1,323	708
CTR	11.76%	11.86%
Engagement Rate	13.56%	13.60%
Conversion Rate	5.21%	4.82%
Purchase Rate	0.61%	0.57%
Impressions per ad	1,701	1,696

3. **Ad Type Performance** – Compared Carousel, Image, Stories and Video ads across both platforms.

Ad type	Ads	Impressions	Clicks	Purchases	CTR	Conv. Rate	Purchase Rate
Carousel	51	86,673	10,155	521	11.72%	5.13%	0.60%
Image	52	88,164	10,475	489	11.88%	4.67%	0.55%
Stories	64	108,932	12,782	683	11.73%	5.34%	0.63%
Video	33	56,043	6,667	338	11.90%	5.07%	0.60%

4. **Target Gender** – Compared performance by the gender each ad targets.

Target gender	Ads	Impressions	Share	Purchases	CTR	Conv. Rate	Purchase Rate
Female	83	140,946	41.5%	817	11.66%	4.97%	0.58%
Male	46	78,142	23.0%	474	12.03%	5.04%	0.61%
All	71	120,724	35.5%	740	11.80%	5.19%	0.61%

5. **Target Age Group** – Compared performance by the age group each ad targets.

Age group	Ads	Impressions	Share	Purchases	CTR	Conv. Rate	Purchase Rate
18-24	54	91,795	27.0%	577	11.73%	5.36%	0.63%
25-34	46	77,977	22.9%	441	11.92%	4.74%	0.57%
35-44	54	91,500	26.9%	545	11.84%	5.03%	0.60%
All	46	78,540	23.1%	468	11.69%	5.10%	0.60%

6. **Country Performance** – Compared reach and purchase rate by user country, using the 390,956 events (97.7%) that link to a user after ID standardization and the 9,777 users kept in the users table.

Country	Users	Impressions	Share	Purchases	Purchase Rate
United States	2,945	99,920	30.1%	640	0.64%
United Kingdom	1,479	50,255	15.1%	278	0.55%
Canada	983	33,577	10.1%	193	0.57%
India	922	30,997	9.3%	184	0.59%
Germany	813	27,657	8.3%	150	0.54%
Australia	698	23,563	7.1%	121	0.51%
Brazil	589	20,097	6.1%	117	0.58%
Mexico	497	16,982	5.1%	121	0.71%
Japan	482	16,424	4.9%	121	0.74%
France	369	12,663	3.8%	61	0.48%

7. **Monthly Trend** – Compared activity by calendar month (the data covers 7 May to 6 August 2025, so May and August are partial months).

Month (2025)	Impressions	Clicks	Purchases	Purchase Rate
May (from 7 May)	91,293	10,911	538	0.59%
June	112,410	12,905	678	0.60%
July	115,333	13,698	696	0.60%
August (to 6 Aug)	20,776	2,565	119	0.57%

8. **Budget vs. Results** – Grouped the 48 campaigns that have ads ($2,383,052 of the $2,535,924 total budget) into four equal groups by budget size. Allocated budget per purchase is a proxy, not an actual cost per acquisition or ROI.

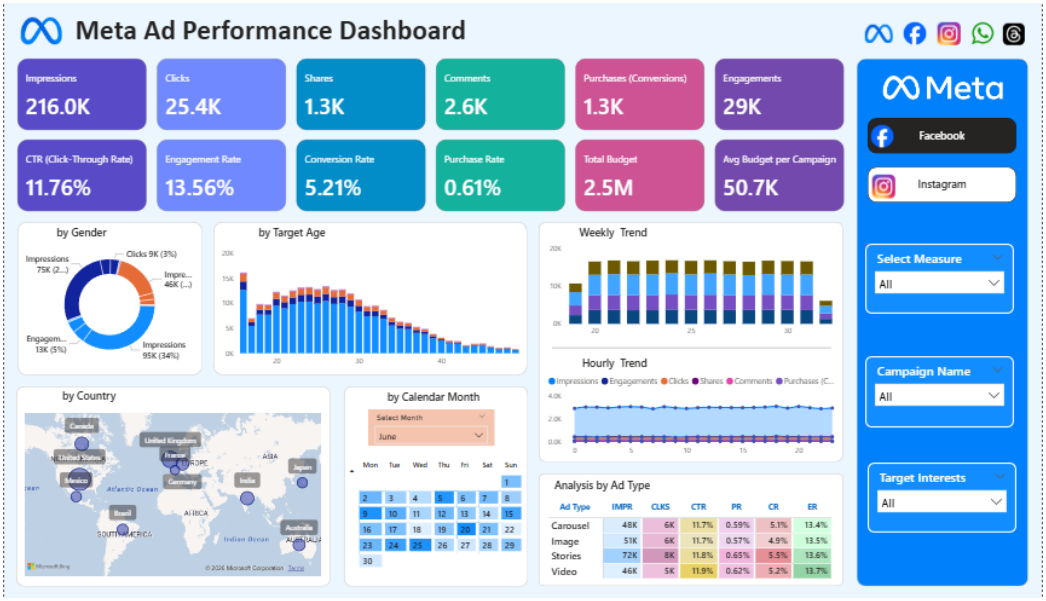
Budget group	Campaigns	Budget ($)	% of budget	Impressions	Purchases	% of purchases	Allocated budget per purchase ($)
Lowest 25%	12	231,393	9.7%	86,228	531	26.1%	436
Second	12	469,540	19.7%	79,927	466	22.9%	1,008
Third	12	695,379	29.2%	88,512	534	26.3%	1,302
Highest 25%	12	986,741	41.4%	85,145	500	24.6%	1,973

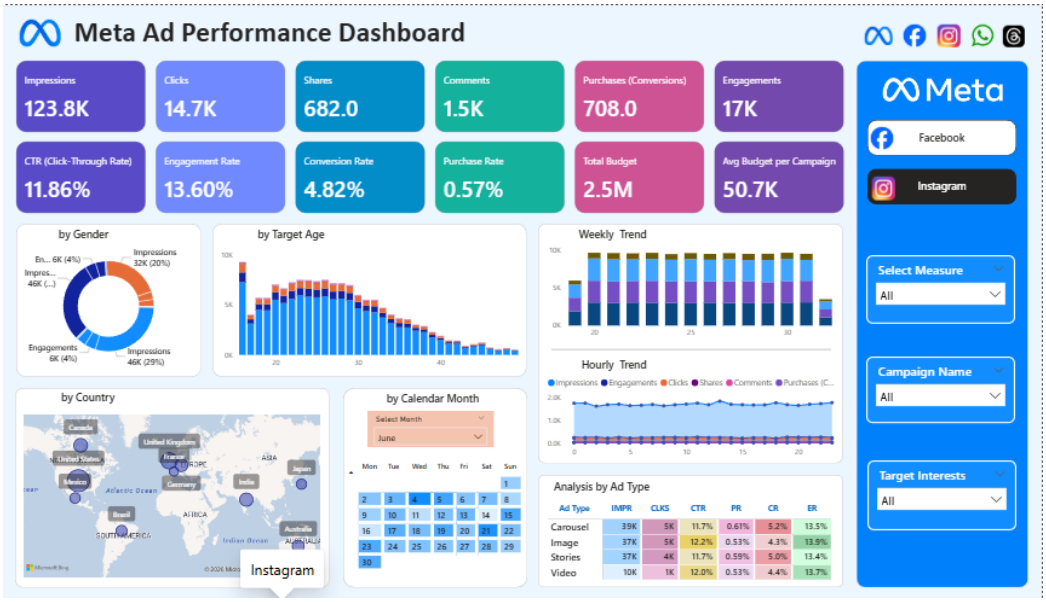
9. **Campaign Efficiency** – Compared the three best and three worst campaigns by allocated budget per purchase.

Campaign	Budget ($)	Ads	Impressions	Purchases	Allocated budget per purchase ($)
Campaign_42_Summer	7,918	8	13,671	67	118
Campaign_29_Winter	19,774	7	11,704	58	341
Campaign_13_Winter	21,855	6	10,331	64	341
Campaign_36_Q3	58,802	2	3,424	10	5,880
Campaign_50_Summer	47,275	1	1,707	8	5,909
Campaign_35_Launch	71,627	1	1,686	11	6,512

5. Dashboard in Power BI

Finally, we built an interactive dashboard in Power BI to present the Meta ad performance insights visually. It can be filtered by platform (Facebook or Instagram), measure, campaign name and target interests, and contains twelve KPI cards and the seven visuals specified in the BRD. The first view below has Facebook selected and the second has Instagram selected.





Independent KPI Validation – Power BI vs. Analytical Dataset

The KPI measures implemented in Power BI were independently validated against the calculated results from the prepared analytical dataset. This reconciliation verifies the accuracy of DAX calculations, aggregation logic and dashboard outputs. Dashboard values are displayed using rounded formatting, while the analytical dataset contains the corresponding exact values.

KPI	Facebook – Power BI	Facebook – Analytical Dataset	Instagram – Power BI	Instagram – Analytical Dataset
Impressions	216.0K	215,972	123.8K	123,840
Clicks	25.4K	25,389	14.7K	14,690
Shares	1.3K	1,275	682	682
Comments	2.6K	2,632	1.5K	1,476
Purchases	1.3K	1,323	708	708
Engagements	29K	29,296	17K	16,848
CTR	11.76%	11.76%	11.86%	11.86%
Engagement Rate	13.56%	13.56%	13.60%	13.60%
Conversion Rate	5.21%	5.21%	4.82%	4.82%
Purchase Rate	0.61%	0.61%	0.57%	0.57%
Total Budget	2.5M	$2,535,924	2.5M	$2,535,924
Avg Budget per Campaign	50.7K	$50,718	50.7K	$50,718

6. Key Findings

- **Platforms Perform the Same** – Neither platform has a clear performance advantage. CTR is 11.76% vs. 11.86% and purchase rate 0.61% vs. 0.57%, and none of the platform differences is statistically significant (CTR p = 0.36, conversion p = 0.09, purchase rate p = 0.14).

- **Reach Follows Ad Count** – Every one of the 200 ads receives between 1,520 and 1,815 impressions (about 1,700 on average), so total impressions mainly reflect how many ads a segment has: Facebook (64% of ads and impressions), Stories, Female-targeted, and 18-24 and 35-44 ads.

- **Budget Is Not Buying Results** – Budget is only weakly related to impressions (r = 0.13) and purchases (r = 0.11). The highest-budget quarter of campaigns uses 41% of spend for 25% of purchases ($1,973 vs. $436 allocated budget per purchase in the lowest group), and the weakest campaigns pair large budgets with only one or two ads.

- **Country Is the Real Differentiator** – Purchase rate differs significantly by country (p = 0.011). Japan (0.74%), Mexico (0.71%) and the United States (0.64%) convert best; France (0.48%) and Australia (0.51%) worst. France rests on only 61 purchases, so it should be confirmed.

- **No Timing Pattern** – Full weeks each bring about 26,000 impressions, hourly impressions range only from 13,971 to 14,346, and weekdays from 48,426 to 48,759. Purchase rate by weekday (0.55% to 0.65%) rests on only 268–314 purchases per day and is likely noise.

- **Formats and Audiences** – Stories has the strongest conversion (5.34%) and purchase rate (0.63%), and Facebook Stories is highest of all (0.65%). Instagram Image has the highest CTR (12.17%) but the lowest conversion (4.35%), and Instagram Video has only 6 ads. Male-targeted ads have a slightly higher CTR (12.03% vs. 11.66% for Female, $p = 0.03$); other ad type, gender and age differences are not significant.

- **Data Notes** – The data holds no revenue and no spend by platform or ad, so ROI cannot be calculated; allocated budget per purchase is only a campaign-level proxy, not a cost per acquisition. Total Budget stays at $2.5M on both platform views because budgets are stored per campaign and 36 of 48 campaigns with ads run on both platforms. The event counts look sampled or simulated (unusually uniform), May and August are partial months, and 9,044 events (2.3%) could not be linked to a user.

7. Business Recommendations

- **Hold Platform Budgets** – Facebook and Instagram perform the same on every rate. Run a controlled test with comparable numbers of ads on each platform and record spend by platform before reallocating budget.

- **Link Budgets to Results** – Review the 12 highest-budget campaigns ($1,973 allocated budget per purchase), cap budgets on campaigns running only one or two ads (Campaign_35_Launch, Campaign_50_Summer and Campaign_36_Q3 exceed $5,800 per purchase), and use Campaign_42_Summer, Campaign_29_Winter and Campaign_13_Winter as benchmarks.

- **Release Unused Budget** – Campaigns 16 and 43 hold $152,872 (6.0% of budget) but run no ads. Confirm whether they are planned, cancelled or mis-recorded.

- **Test Stories and Fix Instagram Image** – Add Stories ads on both platforms to confirm their higher purchase rate, review the landing experience for Instagram Image (clicks convert least), and add Instagram Video ads before judging the format.

- **Prioritize Markets by Purchase Rate** – Increase investment in Japan, Mexico and the United States, and investigate France and Australia before cutting spend.

- **Skip Day-Parting and Fine Audience Splits** – Timing, age and gender show little difference. If one audience test is wanted, add more Male-targeted ads to see whether the small CTR advantage holds.

- **Improve the Tracking Data** – Capture spend and revenue by platform and ad so ROI can be measured, export user IDs as text at the source so the standardization step is no longer needed, and align campaign dates with event timestamps.